Nicholas B. Pratt

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Education

University at Buffalo, Buffalo, NY — Expected May 2026
Bioinformatics & Computational Biology B.S.
School of Engineering and Applied Sciences

Research Experience

Undergraduate Research Assistant — 01/26 – Present
Principal Investigator: Zhong Wang; Dr. Jaroslaw Zola
SCoRe (Scalable Computing Research) Group
Joint Genome Institute, Berkeley National Lab; Department of Computer Science, University at Buffalo
Responsibilities:

- Design and execute experiments to measure quality of genomic data for training of genomic foundation models using information theory metrics.

Laboratory Technician — 08/25 – 01/26
Principle Investigator: Dr. Charlotte Lindqvist
Department of Biological Sciences, University at Buffalo

- Paid research position working in close collaboration with the Albert Lab on S. purpurea & U. gibba projects.
- Work on nanopore sequencing, genome assembly, functional annotation, construction of evolutionary history, and characterization of genome architecture for various plant and bear species.

Summer/Undergraduate Research Assistant — 06/25 – 12/25
Principal Investigator(s): Dr. Michael Gore, Dr. Kelly Robbins
Plant Breeding and Genetics Section, Cornell University, Boyce Thompson Institute
Summer Program: CROPPS Programmable Plant Systems REU

- Construct and preprocess datasets from 2023 and 2024 field trials of a prototype programable plant system in maize with mutant *slac1* genes
- Develop and implement a longitudinal modeling pipeline combining Gaussian curve fitting, linear mixed models, and Random Forest analysis in R to identify genomic and environmental drivers of variation in potential tolerance to drought

Undergraduate Research Assistant — 09/24 – Present
Principle Investigator: Dr. Victor Albert
Department of Biological Sciences, University at Buffalo
Responsibilities:

- Assembled and scaffolded the first draft genome of Sarracenia purpurea using long-read assembly pipelines and Hi-C chromosomal interaction data, followed by quality assessment (BUSCO, QUAST) and iterative polishing.
- Performed functional annotation and comparative genomic analyses, identifying digestive enzyme gene families and evaluating patterns of convergent evolution

Summer Research Assistant 06/24 – 08/24
Principal Investigator: Dr. Trevor Krabbenhoft
Department of Biological Sciences, University at Buffalo
Summer Program: SUNY Chancellor Biological Sciences REU
Responsibilities:

- Retrain publicly available machine learning models to create a tool to detect Oxidative DNA Damage during DNA sequencing pipeline.
- Perform using Oxford Nanopore sequencing pipelines. Protocol includes DNA extraction, library preparation, nanopore sequencing, and sequence basecalling using bioinformatic tools.

Undergraduate Research Assistant 01/24 – 12/24
Principal Investigator: Dr. Jaroslaw Zola
SCoRe (Scalable Computing Research) Group
Department of Computer Science, University at Buffalo
Responsibilities:

- Work on improving metagenomic classifier Coriolis and provide in depth analysis of how changing parameters of the tool effect throughput and accuracy.
- Statistical analysis measuring accuracy, precision and recall of Coriolis compared to other metagenomic classifiers to quantify the effectiveness of the Coriolis algorithm.

Publications

Albert, V. A., Kirshner, J., Page, C., **Pratt, N.**, Merkel, J., Richter, M., Freund, M., Fukushima, K., & Lindqvist, C. Genome evolution and convergent innovation in the carnivorous plant *Sarracenia purpurea*. *bioRxiv*. 2025. doi: 10.64898/2025.12.26.696377

Honors

UB Excellence in Research, Scholarship and Creativity Award 2025

- University-wide award recognizing one undergraduate research project per academic division for exceptional achievement in content, scope, and presentation.

Barry M. Goldwater Scholarship 2025

- Prestigious national scholarship recognizing outstanding undergraduate researchers in STEM fields.

Benjamin A. Gilman International Scholarship 2024

- U.S. Department of State scholarship that provides funding for undergraduate students with financial need to study or intern abroad, supporting global engagement and cross-cultural understanding. Studied abroad in Czechia in 2024.

McNair Scholar Program, University at Buffalo 2024 – Present

- Ronald E. McNair Post-Baccalaureate Achievement Program is dedicated to excellence in scholarship. Serves low-income, first-generation students and traditionally underrepresented undergraduate students along their path to earning a PhD.

University at Buffalo (UB) Dean's List 2022 – Present

- Full-time undergraduate students who demonstrate academic excellence by earning a grade point average (GPA) of at least 3.6.

Daniel Acker Scholars Program & Scholarship, University at Buffalo 2022 – Present

- The Daniel Acker Scholars Program is a unique, individualized learning community of academically talented and diverse students who have strong commitment to justice, leadership and community service.

NYS Triple C Award 2022

- The Triple C Award issued by the *New York State Attorney General's Office* for having an inspiring character, courage and commitment (to community service or leadership) in the face of personal adversity and life challenges. The award recognizes student's spirit and determination.

Presentations

Pratt, N. (2025, October). *Longitudinal Modeling of Maize Gas Exchange for Optimizing Water Use Efficiency*. Oral presentation at the Gulf Coast Undergraduate Research Symposium, Houston, TX.

Pratt, N. (2025, August). *Genomic and Environmental Effects on Maize Gas Exchange*. Oral presentation at the Boyce Thompson Institute George and Helen Kohut Research Symposium, Ithaca, NY.

Pratt, N., Page, C. (2025, April). *Sarracenia purpurea: Genome Assembly and Analysis of the Carnivorous North American Purple Pitcher Plant*. Poster presentation at the Celebration of student excellence Research Showcase, Buffalo, NY.

Pratt, N. (2025, April). *Enhancing Nanopore Sequencing: Leveraging Machine Learning to Detect Oxidative DNA Lesions*. Poster presentation at the 2025 SUNY AI Symposium at UB, Buffalo, NY.

Pratt, N. (2025, March). *Sarracenia purpurea: Hierarchical Convergence in Morphology and Genome Evolution of Carnivorous Pitcher Plants*. Poster presentation at the 26th Annual McNair Scholars Undergraduate Research Conference, College Park, MD.

Pratt, N. (2024, November). *Enhancing Nanopore Sequencing: Leveraging Machine Learning to Detect Oxidative DNA Damage*. Poster presentation at University at Buffalo First Generation Research Symposium, Buffalo, NY.

Pratt, N. (2024, July). *Enhancing Nanopore Sequencing: Leveraging Machine Learning to Detect Oxidative DNA Lesions*. Poster presentation at the SUNY Chancellor's Biological Sciences Summer Undergraduate Research Excellence Program Symposium, Buffalo, NY.

Work Experience

Resident Assistant 08/25 – Present

University at Buffalo Campus Living, Buffalo, NY

- Support 40 residents by fostering a safe and academically focused living environment.
- Plan and lead community programming, mediate conflicts, and connect residents with university resources.
- Provide ongoing peer mentorship and help new students adjust to academic and campus life.

Project Manager/Estimator/Laborer 06/23 – Present
The MJA Company, Buffalo, NY

- Did and estimate price construction projects, developed standardized proposal systems, coordinate crews and schedules, and ensure execution matches project plans and budgets.
- Work independently installing epoxy flooring and chemical floor penetrants, polishing and repair concrete flooring.

Grocery Clerk/Maintenance Personnel 05/19 – 06/23
Various Employers, Buffalo, NY

- Worked in multiple seasonal and part-time roles including landscaping, agricultural labor, retail assistance, housekeeping, and other service positions.
- Developed transferable skills in time management, communication, and problem solving.

Media Appearances

April 2025 University at Buffalo News Center. "Two Buffalo students awarded Goldwater scholarships." Link

Certifications

- OHSA 10 Certification
- Laboratory Safety Certification

Professional Activities

UB Chemistry Club
Land Restoration Project
Founder/Vice President of UB Graphic Design Club
WNY Youth Climate Council
Blood Drive Student Organizer

Service

WNY Grassroots Gardens - Volunteer	06/23 – 08/23
St. Luke's Mission of Mercy, Buffalo, NY - Volunteer	09/18 – 11/22
The Genesis Center of Western New York, Buffalo, NY - Volunteer	06/16 - 07/18